COMSATS University Islamabad, Attock Campus

Department of Computer Science

Program: BAI

Fall 2025: Lab Project	Course: Object Oriented Programming (CSC241)	
Dated: 09/09/2025	Due Dated: Final Exam	Marks: 100

Lab Project Description

Objective:

Work in teams to design and implement a GUI-based application that demonstrates your understanding of key Object-Oriented Programming (OOP) concepts: including classes, inheritance, polymorphism, encapsulation, file handling, and event-driven programming.

- Each team should select a real-world problem domain and develop a desktop-based solution using Java Swing for GUI implementation. Projects should emphasize modularity, reusability, and code clarity.

Topics to be Covered in the Project:

1. Classes, Encapsulation, and Object Relationships

- Define classes with appropriate data members and member functions.
- Demonstrate encapsulation through privacy, visibility modifiers, and getter/setter methods.
- Implement constructors, constructor chaining, static data members, finalizers, and proper memory management.
- Illustrate Composition and Aggregation (Has–A Relationship) to model real-world dependencies.
- Demonstrate method overloading for compile-time polymorphism.

2. Inheritance and Method Overriding

- Use inheritance to create subclasses and hierarchical relationships.
- Apply method overriding to redefine behavior in derived classes.
- Maintain encapsulation, abstraction, and code reusability.

3. Sub-Typing and Polymorphism

- Implement dynamic dispatch for method calls.
- Demonstrate implicit up-casting and explicit down-casting.
- Explore behavioral replacement (subtypes acting like supertypes).

- Utilize abstract base classes and interfaces.
- Employ generic types, static, and dynamic typing concepts.

4. Library Components and File Handling

- Use collection classes and iterators for efficient data handling.
- Apply file I/O operations to read and write structured data.
- Demonstrate exception handling and input validation.

5. Graphical User Interfaces (GUI) and Event Handling

- Use Swing components to design interactive interfaces.
- Implement event-driven programming using listeners and handlers.
- Demonstrate reactive frameworks and handling of program-generated events.

Suggested Project Domains:

1. Education – Student Management System
2. Healthcare – Patient Record / Appointment System
3. Finance – Expense Tracker / Billing System
4. Productivity – To-Do App / Task Scheduler
5. Transportation – Vehicle Parking System
6. Inventory – Warehouse / Stock Management
7. Library – Book Issuance and Catalog System
8. Hotel Management – Room Booking / Check-in System
9. Retail – Point of Sale (POS) Application
10. HR – Employee Attendance and Payroll System
11. Real Estate – Property Management Application
12. Tourism – Travel Package Management System
13. Banking – Account Management / Loan Tracking
14. Agriculture – Farm Crop Monitoring System
15. Event Management – Registration and Scheduling
16. Restaurant – Order and Billing System
17. Logistics – Parcel / Courier Tracking System
18. Sports – Tournament Scheduling App
19. E-Commerce – Product Catalog and Order Tracking
20. Smart Home – Device Monitoring Dashboard

Submission Guidelines:

- Work in teams of 2–3 students.
- Each project must include a brief documentation report (overview, UML design, and code snippets).
- Source code must be properly indented and commented.
- GUI must be fully functional and demonstrate event-driven programming.
- Plagiarism or direct code copying will result in disqualification.
- Submission Deadline: (To be announced by Instructor)

Evaluation Criteria (Total: 100 Marks):

- Problem Understanding & Requirements (10 marks)
- Object-Oriented Design & UML (20 marks)
- Implementation & Code Quality (30 marks)
- GUI Design & Functionality (20 marks)
- Documentation (10 marks)
- Team Collaboration & Presentation (10 marks)